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Growing from a few cells: combined effects of initial stochasticity and cell-to-cell variability

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The growth of a cell population from a large inoculum appears deterministic, although the division process is stochastic at the single-cell level. Microfluidic observations, however, display wide variations in the growth of small populations. Here we combine theory, simulations and experiments to explore the link between single-cell stochasticity and the growth of a population starting from a small number of individuals. The study yields descriptors of the probability distribution function (PDF) of the population size under three sources of stochasticity: cell-to-cell variability, uncertainty in the number of initial cells and generation-dependent division times. The PDF, rescaled to account for the exponential growth of the population, is found to converge to a stationary distribution. All moments of the PDF grow exponentially with the same growth rate, which depends solely on cell-to-cell variability. The shape of the PDF, however, contains the signature of all sources of stochasticity, and is dominated by the early stages of growth, and not by the cell-to-cell variability. Thus, probabilistic predictions of the growth of bacterial populations can be obtained with implications for both naturally occurring conditions and technological applications of single-cell microfluidics.

1. Introduction

Measuring the growth of a bacterial population is a routine task in microbiology laboratories, where the growth rate is used to evaluate the strain fitness or its response to the environment [1,2]. However, the exponential nature of the population's growth inherently implies that the fittest sub-population dominates the measurements, and that phenotypic variations within the initial inoculum are hidden from view [3]. In this context, the desire to understand the heterogeneity in the population has led to the development of single-cell methods, by leveraging the technological advances in microfluidics and microfabrication, imaging and computing [4–7]. These experimental developments have also been accompanied by new theoretical modelling to explain the observed differences [8–10].

Observing each individual cell, however, remains technically challenging, due to the small size of bacteria and also to the exponential increase in cell numbers, which limits the practical observation times to a small number of generations [11]. Recently, the so-called mother machine has been used to track the growth and division of thousands of individual cells, by discarding the progeny of the mother. This experimental platform has led to the development of important new insights on the role of single-cell stochasticity in phenomena such as the emergence of resistance to antibiotics [12–14], a source of major health concerns [15].

But aside from fundamental biological questions, the impact of microorganisms, e.g. on health or the environment, resides in the behaviour of microbial populations rather than the behaviour of single cells. The growth and evolution of these populations in turn depend on a complex interplay between

single-cell properties, the symbiosis or competition between the cells [16] and the emergence of inheritable mutations [17]. These aspects cannot be observed in the mother machine, since they require the accumulation of individuals in the observation region. In this context, droplet-based methods offer a straightforward way to keep individual cells and their progeny in a closed micro-reactor and to observe their growth in time, while controlling the encapsulation conditions (see [18] for an excellent review). The ability to encapsulate the cells within a closed environment (already used by Lwoff in the 1940s [19]) also allows studies of bacterial interactions such as quorum sensing [20,21] or competition between different strains [22].

Below we present a theoretical framework to map the stochasticity of the single-cell behaviour onto probabilistic predictions on the scale of the population. The theory is based on the classical Bellman–Harris (B-H) model [23], which we extend to include the different sources of uncertainty of the microfluidic experiments: cell-to-cell variability, the Poisson distribution of the initial number of cells per droplet, and the change of the cell division rates with the generation number of the cell. The model's analytical predictions are compared with distributions obtained from Monte Carlo simulations and the two agree very well for all of the situations studied. The two approaches thus indicate how the different measurable parameters depend on each of the sources of stochasticity. In particular, we study the probability distribution function (PDF) of the population size. We show that all moments of the PDF grow exponentially, with the same growth rate, which depends solely on cell-to-cell variability. In contrast, the shape of the PDF is predominantly determined by early stochastic events, namely the initial distribution of cells per droplet and the generation-dependent rate of cell division. Then the analytical predictions are compared with experimental measurements in microfluidic droplets. By measuring the growth of populations in hundreds of parallel droplets, the microfluidics act as *experimental Monte Carlo* simulations, in which the distribution of observed behaviours integrates the accumulation of stochastic effects in each droplet. The comparison between theory and experiments displays very good quantitative agreement for the exponential phase of growth, indicating that the model captures all of the essential features of the experimental conditions.

2. Theoretical and numerical results

Bacterial division and growth at the single-cell level have been described by three different models: (i) the 'timer' model, in which the cells divide once they have reached a given age [24]; (ii) the 'sizer' model that supposes that the cells divide once they have reached a given size [25]; (iii) the 'adder' model in which the size increase between two divisions is conserved for all cells [8]. However, it is still an active topic of research to know how these different single-cell models determine the growth at the population scale [10]. Here, rather than considering the behaviour of each cell in balanced growth, we wish to understand how this single-cell behaviour affects the growth of a population: if we encapsulate bacteria in a thousand isolated micro-droplets, and let them divide over a few hours according to the rules of each of the single-cell models, will the distributions of number of bacteria be different for the three models? Note that bacteria are encapsulated randomly in

droplets, so that the initial number of bacteria per droplet follows a Poisson distribution [26]. With our experimental constraints (≈ 1000 droplets), we often choose a Poisson parameter close to 1. This guarantees a sufficient number of droplets containing bacteria for a meaningful statistical analysis, as well as a low enough initial number of bacteria per droplet to keep a trace of the cell-to-cell variability.

2.1. Comparing the three models of cell division

In all three models, the volume (or, equivalently, the length) of individual bacteria grows exponentially with a rate μ . The difference between the three models lies in the criterion for cell division, as described above. The three models are stochastic to account for the individual variability of bacterial cells [3]. There are two sources of noise, both of which are modelled by Gaussian variations: one on the division time, $\xi \sim \mathcal{N}(0, \sigma_\xi)$, and one on the elongation rate $\mu \sim \mathcal{N}(\mu_0, \sigma_\mu)$. These parameters depend on the bacterial strain and the environment in which it grows. For instance, the classical 'growth law' [27] relates the mean growth rate μ_0 to the composition of the medium. All three models can be summarized in two equations for the volume of the cell at division v_d and for the division time τ_d [9,10]:

$$v_d = 2pv_0 + 2(1-p)v_b \quad (2.1)$$

and

$$\tau_d = \frac{1}{\mu} \ln\left(\frac{v_d}{v_b}\right) + \xi, \quad (2.2)$$

where v_b is the volume at birth and v_0 is the constant volume after which division takes place in the adder model. In the equations above, the value of the parameter p defines the regulation strategy of a cell: for $p = 0, 0.5, 1$, the cell follows a timer, adder or sizer model, respectively. Indeed, if $p = 0$, then $v_d = 2v_b$, and by substituting in equation (2.2), $\tau_d = \ln(2)/\mu + \xi$. The division time of the cells is thus fixed at $\ln(2)/\mu$ plus a stochastic noise ξ , which is a timer model. If $p = 0.5$, then $v_d = v_0 + v_b$ so that all cells add a volume v_0 to their original size v_b before dividing (adder model). Last, if $p = 1$, then $v_d = 2v_0$, so that the volume at division is constant (sizer model).

At the population level, all models lead to an exponentially growing number of bacteria $N(t) \sim e^{\alpha_p t}$, where the growth rate α_p depends on the model. In the case of weak noises ($\mu_0 \sigma_\xi \ll 1$), the growth rates of the adder and sizer models are the same, while the growth rate for the timer model is very similar to the one of the other two models, with a difference smaller than 4% [10]. Such small differences are extremely hard, if not impossible, to detect experimentally on measurements at the population scale.

In the general case of arbitrary noise, analytical predictions are difficult to make and one has to rely on numerical simulations to know the growth behaviour of the population. Here, we carry out Monte Carlo simulations mimicking the microfluidic experiment: identical simulations of bacterial growth are run in parallel 1000 times, with the initial number of cells n_0 being chosen according to a Poisson distribution: $P_\lambda(N_0) = \lambda^{N_0} e^{-\lambda}/N_0!$ with a Poisson parameter $\lambda = 0.75$. We choose $\sigma_\xi \sim 0.25 \ln(2)/\mu_0$ and $\sigma_\mu = 7\%$, in accordance with our experimental values and those of [8]. The three models yield very similar distributions of the number of cells as a function of time (figure 1).

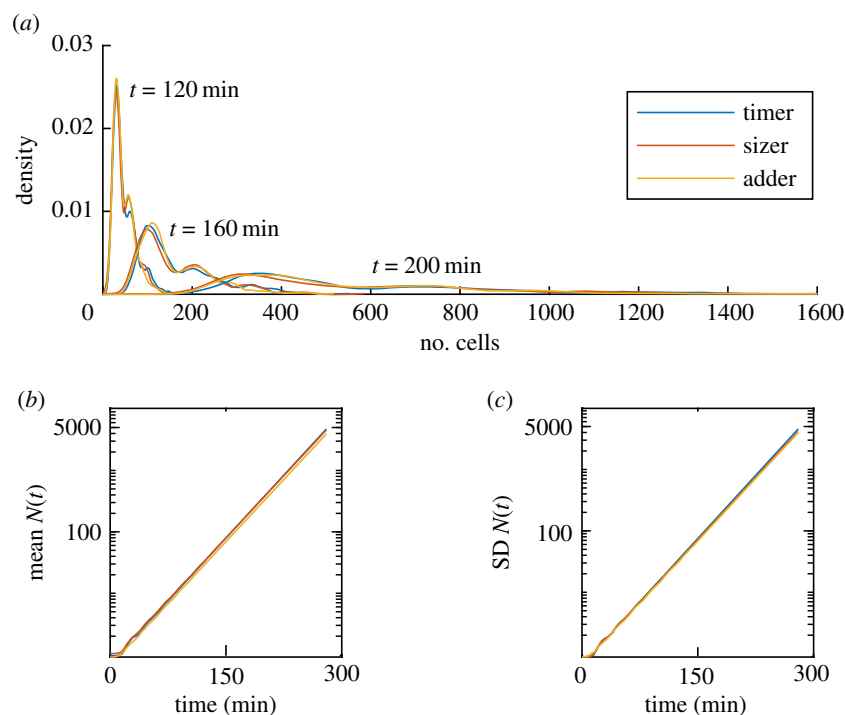


Figure 1. Comparison between the three models: timer, adder and sizer, with an initial Poisson distribution. $\sigma_\xi = 0.2 \ln(2)/\mu_0$, $\mu_0 = 0.031 \text{ min}^{-1}$ ($\tau_0 \sim 22 \text{ min}$), $\sigma_\mu = 0.1\mu_0$, number of realizations=1000, $\lambda = 0.75$. (a) Distributions of the number of cells at $t = 120$, 160 and 200 min, respectively. (b) Mean number of bacteria and (c) standard deviation of the number of bacteria per simulation, in log scale. (Online version in colour.)

For further comparison with the microfluidic experiments, we run a complete set of simulations covering the range of experimental values of σ_μ and σ_ξ found by Taheri *et al.* [8]: $0.1 \leq \sigma_\xi \leq 0.3$, and $0.025 \leq \sigma_\mu \leq 0.15$. The initial number of bacteria is sampled from a Poisson distribution with a Poisson parameter $\lambda = 0.75$. Bacteria are allowed to grow and divide for approximately 4 h, a time close to the duration of the exponential phase in our experiments. The final distributions of the number of bacteria, rescaled by their mean, are compared using a two-sampled Kolmogorov–Smirnov test [28] to know whether the outputs of the three models are distinguishable. For each point of the $[\sigma_\mu, \sigma_\xi]$ mesh and each of the three models, we find that there always exists a distribution belonging to another model that is statistically indistinguishable from it at a 5% significance level, as shown in electronic supplementary material, figure S1. For experimental purposes, the three division models are therefore indistinguishable at the population level and give the same results. Therefore, any model can be used to predict the growth of a population. In the following, we extend the classical timer model of Bellman & Harris [23] to describe a larger set of experimental conditions, which allows us to obtain analytical predictions on the population behaviour.

2.2. A generalized Bellman–Harris model

2.2.1. Classical case studied by Bellman and Harris

The timer model was studied by Bellman and Harris in their classic paper [23]. Their original assumption was that the division times of all bacteria are independent and picked from a continuous stochastic distribution which is constant over time. Starting from one cell at $t = 0$, the B-H model derives several properties of the probability distribution of the number of cells as a function of time $N(t)$. For instance, it

shows that all normalized moments $\bar{\mu}_k$ of $N(t)$ grow exponentially with the same growth rate α , and are equivalent to $n_k e^{\alpha t}$ when t goes to infinity, which we will write as $\bar{\mu}_k \sim n_k e^{\alpha t}$ in what follows (see electronic supplementary material for precise definition).

Here we consider that bacteria divide after a time τ_d , which is normally distributed around a mean value τ_0 with standard deviation σ . This distribution allows us to define a microscopic coefficient of variation $cv_\mu = \sigma/\tau_0$ that describes the variability between individual cells. In this case, the value of the growth rate α can be related to τ_0 and cv_μ (see electronic supplementary material for derivation) as

$$\alpha = \frac{\ln(2)}{\tau_0} \frac{2}{1 + \sqrt{1 - 2 \ln(2) cv_\mu^2}}. \quad (2.3)$$

The growth rate is therefore a product of two terms: the first term is the ratio $\ln(2)/\tau_0$, which is the growth rate when the heterogeneity of division times is neglected. The second term is larger than one and increases with cv_μ . It predicts that a larger cell-to-cell variability of the division times increases the apparent growth rate of the whole population, which was confirmed experimentally by Hashimoto *et al.* [29]. Although this dependence of α on cv_μ is verified numerically as shown in figure 2a, the net effect on the growth rate is weak, remaining below 5% for any value of cv_μ within a realistic range.

These observations are confirmed by Monte Carlo simulations, the results of which are in excellent agreement with the analytical prediction of equation (2.3), thus validating the theoretical result, as shown in figure 2a. The divergence that is observed between the numerical and theoretical results for $cv_\mu > 0.3$ is due to the increased apparition of negative division times in the theory, which are removed from the simulations.

It is also interesting to obtain the prefactors n_1 and n_2 of the mean (M_N) and standard deviation (SD_N) of the

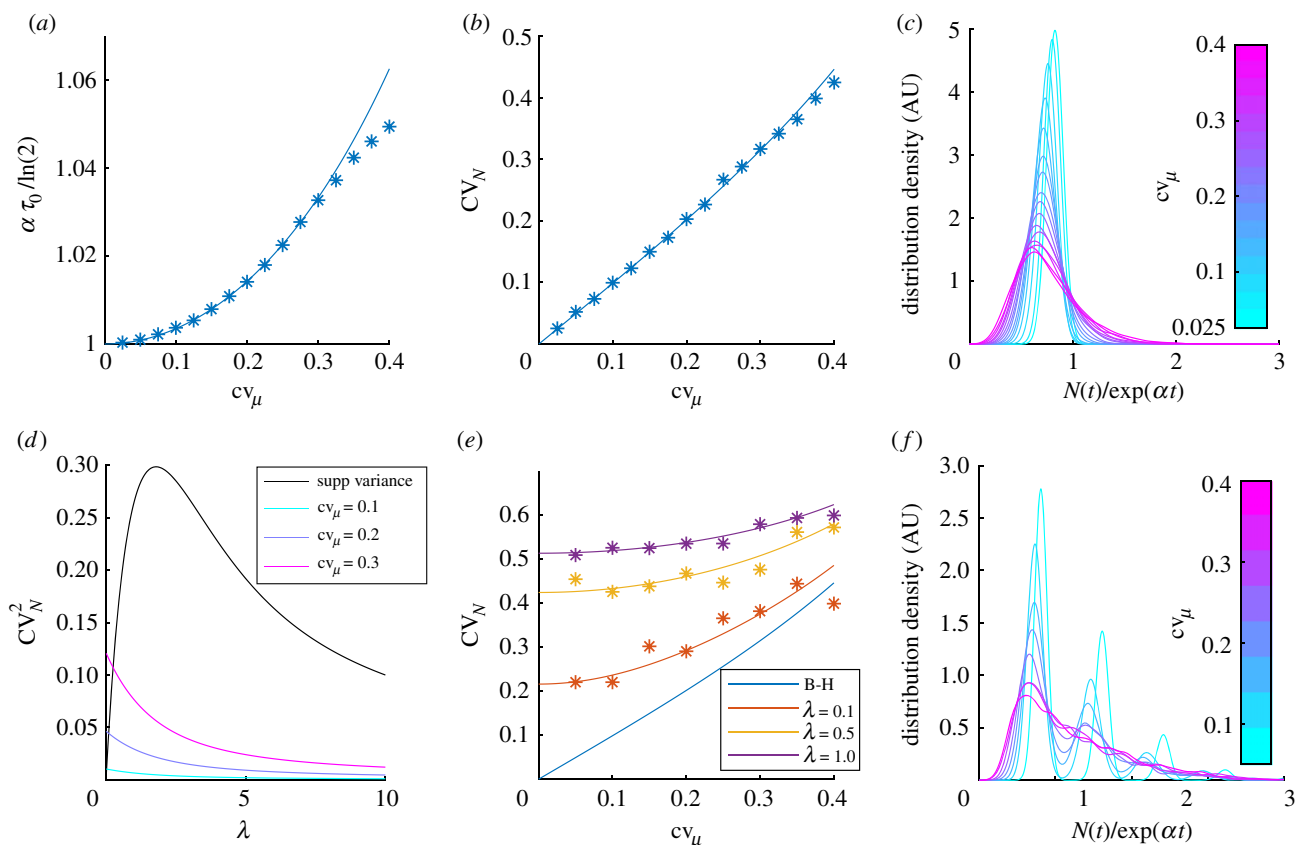


Figure 2. (a–c) Theory and simulations for the B-H model with a Gaussian distribution for the division times. Simulations were made with $\tau_0 = 21$ min, 2000 simulations going from $t = 0$ to $t = 5$ h. (a,b) Theory (lines) and simulations (stars). (a) Gain in the exponential growth rate as a function of the cellular-level coefficient of variation cv_μ . (b) Macroscopic coefficient of variation CV_N as a function of cv_μ . (c) Asymptotic shape of the distribution of $N(t)/\exp(\alpha t)$. The curves show a kernel fit of the simulated curves. (d–f) Theory and simulations for the B-H model with a Gaussian distribution for the division times and a Poisson distribution for the initial number of cells. (d) Contributions of the B-H process (coloured lines, term (1) in equation (2.12)) and of the initial Poisson distribution (black line, term (2) in equation (2.12)) to the square of CV_N , as a function of the parameter of the Poisson distribution and for different values of cv_μ . (e) CV_N , asymptotic value coefficient of variation of the number of cells as a function of cv_μ , theory (lines) and simulations (stars). (f) Asymptotic shape of the distribution of $N(t)/\exp(\alpha t)$ (kernel fit, simulations only), for $\lambda = 1$. (Online version in colour.)

distribution, respectively. The expressions of n_1 and n_2 are functions of τ_0 and σ , the parameters of the Gaussian law for single cell division, as derived in the electronic supplementary material, section S2. Then the *macroscopic* coefficient of variation, defined as $CV_N(t) = SD_N(t)/M_N(t)$ asymptotically converges to a constant value $CV_N(\infty) = n_2/n_1$ that can be expressed as follows (see the electronic supplementary material for derivation):

$$CV_N^2(\infty) = \left(\frac{n_2}{n_1}\right)_{\text{BH}}^2 = \frac{e^{\alpha^2 \sigma^2} - 1}{1 - (e^{\alpha^2 \sigma^2}/2)}. \quad (2.4)$$

This expression depends strongly on cv_μ , as shown in figure 2b. Therefore, although the cell-to-cell variability has only a small effect on the total growth rate, its signature is clear on the macroscopic coefficient of variation between different realizations. Again, the theoretical prediction is validated by the numerical Monte Carlo simulations.

Another consequence of the B-H model is that the asymptotic distribution of the number of cells, when rescaled by $\exp(\alpha t)$, converges to a stationary distribution. This prediction is also recovered by the numerical simulations, as shown in electronic supplementary material, movie S1, where the distribution of $N(t)/\exp(\alpha t)$ is seen to converge to a constant curve at late times. Similarly to CV_N ,

the shape of this distribution depends strongly on the single-cell parameters, as plotted in figure 2c.

Taken together, the above results show the effects of the cell-to-cell variability on the population-scale growth. While the impact on the exponent α is always weak, the shape of the asymptotic probability distribution of $N(t)$ is very sensitive to the value of cv_μ . This observation suggests that the variability between cells can be inferred from a large number of population-scale measurements without the need for single-cell resolution. However taking into account other sources of variability in the practical situations will limit such inference, as demonstrated in the following sections.

2.2.2. Poisson distributed initial number of cells

When bacteria are encapsulated in droplets, the initial number of bacteria per droplet follows a Poisson distribution of parameter λ [30]. The probability of having N_0 bacteria in a droplet is given by: $P_\lambda(N_0) = \lambda^{N_0} e^{-\lambda}/N_0!$ [26], where λ is the average number of bacteria per droplet. In a microfluidic set-up, the Poisson distribution leads to some droplets being empty, but we solely consider the growth in droplets containing bacteria. As bacteria are assumed to be independent from one another, this modification of the initial number of bacteria does not change the asymptotic behaviour of the distribution: the k th moment of the distribution of $N(t)$ grows exponentially

with the same growth rate α as previously, but the values of the prefactors in front of the exponential are different from the classical case.

To derive the values of the prefactors, consider a droplet containing initially N_0 bacteria, with $N_0 > 0$. We call $N_k(t)$ the size of the offspring of bacteria k at time t . The total number of bacteria $N_\lambda(t)$ in the droplet at time t is then given by

$$N_\lambda(t) = \sum_{k=1}^{N_0} N_k(t). \quad (2.5)$$

Since we suppose that bacteria grow independently from one another, the terms of the sum are independent and identically distributed. Thus, the probability generating function $G_t(s)$ of $N_\lambda(t)$, which can be defined as $G_t(s) = \mathbb{E}(s^{N_\lambda(t)})$, can be obtained as the composition of the probability generating function of the classical B-H model and of the probability generating function of the Poisson distribution restricted to the positive integers

$$G_t(s) = g_\lambda[g_t(s)]. \quad (2.6)$$

Here $g_t(s)$ is the probability generating function of the number of bacteria $N_k(t)$ for the classical B-H model starting with exactly one initial bacterium, and g_λ is the probability generating function for the random initial number of bacteria n_λ .

We know that n_λ follows a Poisson distribution, so

$$g_\lambda(s) = \frac{e^{-\lambda}}{1 - e^{-\lambda}} (e^{\lambda s} - 1). \quad (2.7)$$

We do not know the expression for $g_t(s)$, but we know the values of its first and second derivatives in $s = 1$: they are the first two moments of the distribution in the classical B-H case. We can therefore compute the first and second derivatives of $G_t(s)$ and find the mean and variance of $N_\lambda(t)$

$$\mathbb{E}_\lambda(t) = \frac{dG_t}{ds} \Big|_{s=1} \quad (2.8)$$

and

$$\text{Var}_\lambda(t) = \frac{d^2 G_t}{ds^2} \Big|_{s=1} + \frac{dG_t}{ds} \Big|_{s=1} - \left(\frac{dG_t}{ds} \Big|_{s=1} \right)^2. \quad (2.9)$$

This leads to a mean $M_\lambda(t) \sim n_1(\lambda)e^{\alpha t}$ with

$$n_1(\lambda) = \frac{\lambda}{1 - e^{-\lambda}} n_1, \quad (2.10)$$

while the standard deviation $\text{SD}_\lambda(t) \sim n_2(\lambda)e^{\alpha t}$ is

$$n_2(\lambda) = \sqrt{\frac{\lambda}{1 - e^{-\lambda}} \left(n_2^2 + \frac{1 - (\lambda + 1)e^{-\lambda}}{1 - e^{-\lambda}} n_1^2 \right)}. \quad (2.11)$$

The expression for $n_1(\lambda)$ is expected because of the independence of the bacterial division times: the mean number of cells at time t is simply the product of the mean number of cells per non-empty droplet and the mean of the classical B-H case. To understand the expression for the standard deviation better, it is useful to divide it by the mean and take the square. Doing so, we obtain the square of the coefficient of variation CV_λ for t going to infinity:

$$\begin{aligned} \text{CV}_\lambda^2(\infty) &= \left(\frac{n_2}{n_1} \right)_\lambda^2 \\ &= \underbrace{\frac{1 - e^{-\lambda}}{\lambda} \left(\frac{n_2}{n_1} \right)_{\text{BH}}^2}_{(1)} + \underbrace{\frac{1 - (\lambda + 1)e^{-\lambda}}{\lambda}}_{(2)}. \end{aligned} \quad (2.12)$$

The variance of the population size is thus the sum of two terms. The first one is the variance due to the stochastic nature of the B-H process, corrected to take into account only the wells that contain bacteria. The second term is an additional variance due to the initial Poisson distribution. The contribution of the initial Poisson distribution of the number of cells becomes greater than the intrinsic variance of the B-H process when the Poisson parameter is larger than ≈ 0.5 , see black line in figure 2*d*. The dominant effect on the variance of the number of cells is then the variance due to the initial distribution, and not the heterogeneity of the division times. This can be seen clearly in figure 2*e*: the coefficient of variation of the population now depends weakly on cv_μ but cases with different values of λ can clearly be distinguished. Simulations confirm these results, and highlight another surprising effect: the numerically computed values tend to differ more from the theoretical prediction for small values than for large values of λ . Indeed, for small values of λ , fewer droplets contain bacteria, and the stochastic divergence from the prediction is bigger, even with 2000 simulations.

Note that, as in the classical B-H case, the rescaled distribution of the number of cells remains convergent for long periods (see electronic supplementary material, movie S2). The shape of the asymptotic distribution now shows peaks centred around the initial number of bacteria in the droplets (figure 2*f*). The width of each peak still depends on the variability of the division times. If we consider only the droplets containing initially k bacteria, and compute the coefficient of variation for the number of bacteria in these droplets, we find $\text{CV}_N(k) = (\text{CV}_N(k=1)/\sqrt{k})$ (see the electronic supplementary material for derivation and figure 2*f*). This means that the impact of cell-to-cell variability on the population variance can still be observed in the presence of an initial Poisson distribution of the number of cells, but only if one is able to count the initial number of bacteria in the droplet, and to sort the droplets according to this number. Moreover, because of the term in $(1/\sqrt{k})$, determining the population variance is only possible for droplets that contain very few bacteria.

2.2.3. Generation-dependent division time

A final ingredient that must be added to the model is to account for the generation-dependent division time of bacteria. Indeed, when bacteria are set to grow in fresh medium the mean and variance of their division time changes with the generation number [31,32]. Once again, we consider that bacteria grow independently from each other, so that modifying the first generation does not change the asymptotic exponential behaviour of the moments, but the prefactors n_1 and n_2 are modified compared to the classical case.

Let τ_i and σ_i be the mean division time and its standard deviation for the i th generation, where division times are always picked from a Gaussian distribution. To understand how to derive the prefactors, let us study the simpler case where only the first-generation time is picked from a different distribution than the steady state. Let ρ_1 be the density function of the first division time, and ρ_s that of the steady-state division times. The process starts at time $t = 0$ from a single bacterium that divides according to ρ_1 . The two daughter bacteria divide according to ρ_s and give birth to an offspring of size N_1 and N_2 , as shown in figure 3*a*.

Calling t_1 the time at which the initial bacterium has divided, we know that for $t > t_1$, the total number of bacteria

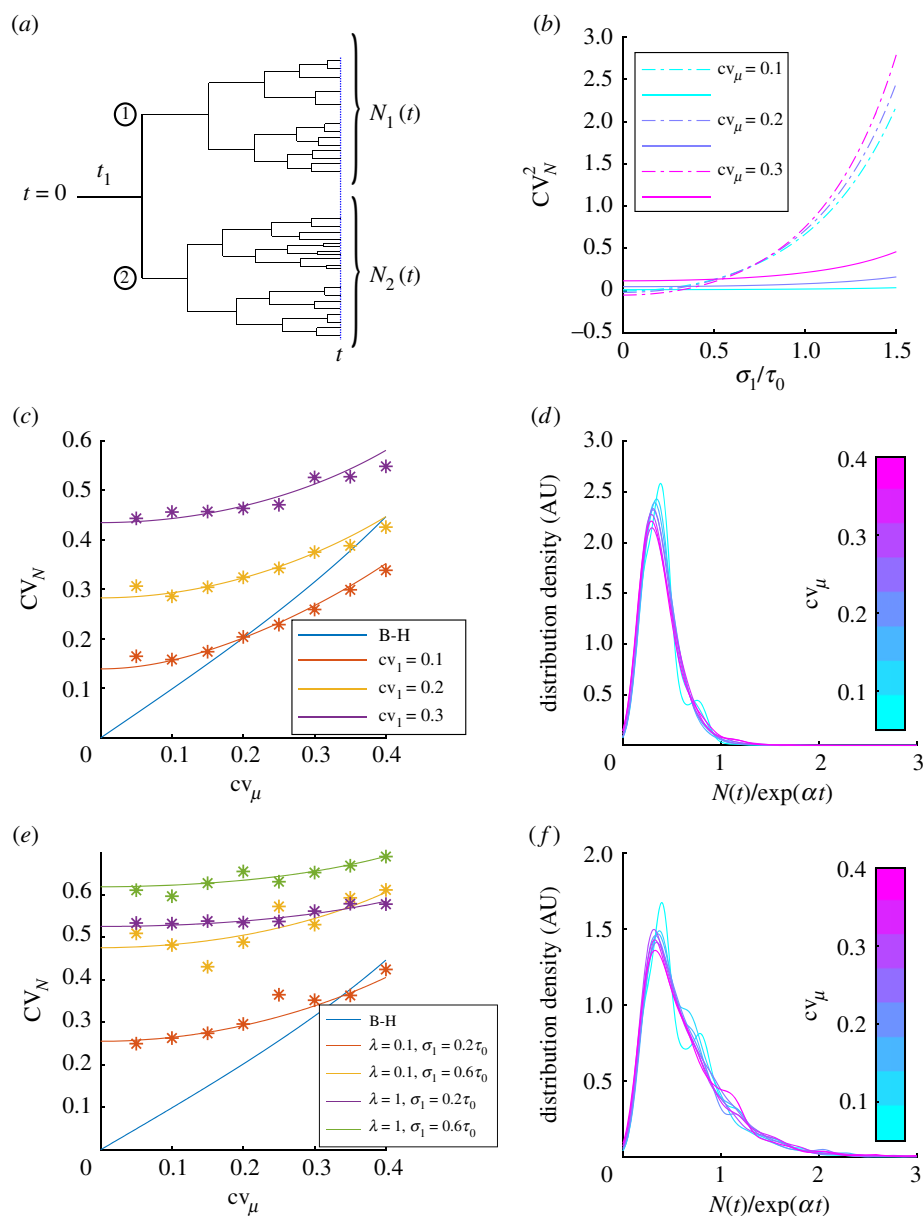


Figure 3. (a–d) B-H model and simulations for a different first generation with $\tau_1 = 2\tau_0$, varying σ_1 . (a) Schematic view of the division scheme in this case: a single bacterium divides into two bacteria according to a division law with density function ρ_1 , with mean τ_1 and standard deviation σ_1 . The two daughter bacteria divide according to ρ_s and give birth to an offspring of size N_1 and N_2 . (b) Theoretical plot of the contributions to CV_N^2 , when varying σ_1 , for different values of cv_μ . Straight lines: contribution of the B-H process, term (1) in equation (2.18). Dotted lines: contribution of the different distribution of division times for the first generation, term (2) in equation (2.18). (c) CV_N , asymptotic value coefficient of variation of the number of cells as a function of cv_μ : theory (lines) and simulations (stars), for different values of $cv_1 = (\sigma_1/\tau_1)$. (d) Shape of the distribution of $N(t)/\exp(\alpha t)$ (kernel fit, simulations only), for $cv_1 = 0.3$. (e, f) Theory and simulations for the mixed case where the first generation is different and the cells are initially distributed following a Poisson distribution. (e) CV_N , asymptotic value coefficient of variation of the number of cells as a function of cv_μ , theory (lines) and simulations (stars). (f) Shape of the distribution of $N(t)/\exp(\alpha t)$ (kernel fit, simulations only), for $\lambda = 1$ and $cv_1 = 0.3$. (Online version in colour.)

at time t is $N(t) = N_1(t - t_1) + N_2(t - t_1)$, where N_1 and N_2 are independent and follow the same law. Therefore, the mean number of bacteria is $\mathbb{E}(N) = \mathbb{E}(N_1) + \mathbb{E}(N_2) = 2\mathbb{E}(N_1)$. After the first division, all daughter bacteria divide according to the classical B-H process. The asymptotic equivalent of $N_1(t - t_1)$ for $t \rightarrow \infty$ is then given by

$$\mathbb{E}(N_1(t - t_1)) \sim n_1 e^{\alpha(t-t_1)}. \quad (2.13)$$

To obtain the expected value $\mathbb{E}(N(t))$ of $N(t)$, we multiply by ρ_1 and integrate over all possible values of t_1

$$\mathbb{E}(N(t)) \sim 2n_1 e^{\alpha t} \int_0^\infty e^{-\alpha t_1} \rho_1(t_1) dt_1. \quad (2.14)$$

Calling $X_1 = e^{-\alpha t_1}$, the mean number of bacteria can be expressed as

$$\mathbb{E}(N(t)) \sim 2n_1 \mathbb{E}(X_1) e^{\alpha t}. \quad (2.15)$$

Similarly, we obtain

$$\mathbb{E}(N_1^2) \sim n_1^2 \frac{2\mathbb{E}(X_1^2)}{1 - 2\mathbb{E}(X_1^2)} e^{2\alpha(t-t_1)}, \quad (2.16)$$

which leads to

$$\mathbb{E}(N(t)^2) \sim 2n_1^2 \frac{\mathbb{E}(X_1^2)}{1 - 2\mathbb{E}(X_1^2)} e^{2\alpha t}. \quad (2.17)$$

From this last result, we obtain the expression of the square of the coefficient of variation in the case where only the first generation is changing, and compare it to the classical B-H expression. The coefficient of variation does not depend on τ_1 but solely on σ_1

$$\begin{aligned} CV_{\sigma_1}^2(\infty) &= \left(\frac{n_2}{n_1}\right)_{\sigma_1}^2 \\ &= \underbrace{e^{\alpha^2(\sigma_1^2 - \sigma^2)} \left(\frac{n_2}{n_1}\right)_{\text{BH}}^2}_{(1)} + \underbrace{e^{\alpha^2(\sigma_1^2 - \sigma^2)} - 1}_{(2)}. \end{aligned} \quad (2.18)$$

This is again the sum of two terms, the first one being directly linked to the intrinsic variance of the B-H process, and the second one being an additional variance due to a first generation time that is different from the following ones. The effect is then similar to the Poisson case: the variance of the distribution of the number of cells is increased when $\sigma_1 > \sigma$, which is generally the case in experiments. When this variance increases, the contribution of the first generation rapidly becomes dominant over the contribution of the B-H process (figure 3b), meaning that the variance of the distribution is mostly due to the early times, and not to the cell-to-cell heterogeneity of division times (figure 3c). Another way to see this result is to look at the shape of the asymptotic rescaled distribution, which is almost independent of the value of the coefficient of variation of the division times (figure 3d).

2.2.4. Putting it all together: three sources of stochasticity

Now what happens when we mix the two effects, i.e. have an initial number of cells per droplet following a Poisson distribution, and the first generation of bacteria having a different generation time? To answer this question, we inject equations (2.17) and (2.15) into equation (2.9). We find that the two

effects add up, leading to the following formula for CV_N^2

$$\begin{aligned} CV_{\sigma_1, \lambda}^2(\infty) &= \left(\frac{n_2}{n_1}\right)_{\sigma_1, \lambda}^2 = \frac{1 - e^{-\lambda}}{\lambda} e^{\alpha^2(\sigma_1^2 - \sigma^2)} \left(\frac{n_2}{n_1}\right)_{\text{BH}}^2 \\ &\quad + \frac{1 - e^{-\lambda}}{\lambda} (e^{\alpha^2(\sigma_1^2 - \sigma^2)} - 1) + \frac{1 - (\lambda + 1)e^{-\lambda}}{\lambda}. \end{aligned} \quad (2.19)$$

We thus have the sum of three terms. The first one corresponds to the contribution due to the variation of the division times of the cells. The second one is the additional variance due to the first division time, but corrected to take into account only the wells that contain bacteria because of the Poisson distribution. And the third one is the supplemental variance due to this Poisson distribution. In this case, the intrinsic variance of the B-H process is now masked by two terms, and as a consequence, the coefficient of variation CV_N is even less dependent on the heterogeneity of the division times cv_μ (figure 3e). The asymptotic shape of the distribution of $N(t)/\exp(\alpha t)$, which is constant in time, is also almost independent of cv_μ (figure 3f).

The same analysis can be extended to the arbitrary case where the first k generations follow different division laws, and combined with an initial number of bacteria following a Poisson distribution. In the particular case of our experiments, the distribution of division times reaches steady state at the fourth generation, and we find that at steady state, the mean $M_{\tau_n, \sigma_n}(t) \sim n_1(\tau_n, \sigma_n)e^{\alpha t}$ and the standard deviation $SD_{\tau_n, \sigma_n}(t) \sim n_2(\tau_n, \sigma_n)e^{\alpha t}$ with

$$n_1(\tau_n, \sigma_n) = 8n_1 \frac{\lambda}{1 - e^{-\lambda}} \prod_{i=1}^3 e^{-\alpha \tau_i + \frac{\alpha^2 \sigma_i^2}{2}}, \quad (2.20)$$

$$CV_{\tau_n, \sigma_n}^2(\infty) = \left(\frac{n_2}{n_1}\right)_{\tau_n, \sigma_n}^2 = \frac{1 - e^{-\lambda}}{\lambda} \left(\frac{1}{8} e^{\alpha^2 \sigma_1^2} \left(e^{\alpha^2 \sigma_2^2} \left(\frac{e^{\alpha^2 \sigma_3^2}}{1 - \frac{1}{2} e^{\alpha^2 \sigma^2}} + 2 \right) + 4 \right) - \frac{\lambda}{1 - e^{-\lambda}} \right). \quad (2.21)$$

Although these formulae are complex, they are simply the reflection of simple ingredients and can easily and reliably be computed for a given set of parameters.

3. Experimental measurements

The mathematical results obtained above can be verified against measurements of bacterial growth, by performing parallel growth experiments in anchored microfluidic droplets. These measurements are complemented by single-cell measurements to provide the parameters for the comparison with the theoretical models.

3.1. Microfluidic experiments

The experiment relies on the recently published ‘universal platform’ for biological assays, in which cells can easily be encapsulated within anchored nanolitre droplets in a microfluidic device [30,33]. The experiments are performed on the chip shown in figure 4a, which consists of a long and

wide chamber whose ceiling is patterned with 1495 cubic holes, which we call ‘anchors’. Briefly, the protocol consists of first priming the microfluidic chip with an inert fluorinated oil, then injecting the bacterial suspension at the correct dilution into the device. The final step, which is generally performed at a well-controlled flow rate, involves pushing oil through the chip again. This second oil flow breaks aqueous droplets in the anchors, where they remain for the duration of the experiment (figure 4b). By having bacteria with culture medium in the droplets, the growth of each population can be followed through time-lapse microscopy, as shown in figure 4c.

The fluorescence signal in each of the wells is used as a proxy for the size of the bacterial population: the fluorescence intensity is proportional to the number of cells in the droplet, with an offset that can be estimated using the signal of empty drops, as discussed in the electronic supplementary material, section S7. An example of the data that can be obtained from these measurements is shown in figure 4d, where we see the increase in fluorescence signal in each of the drops. These

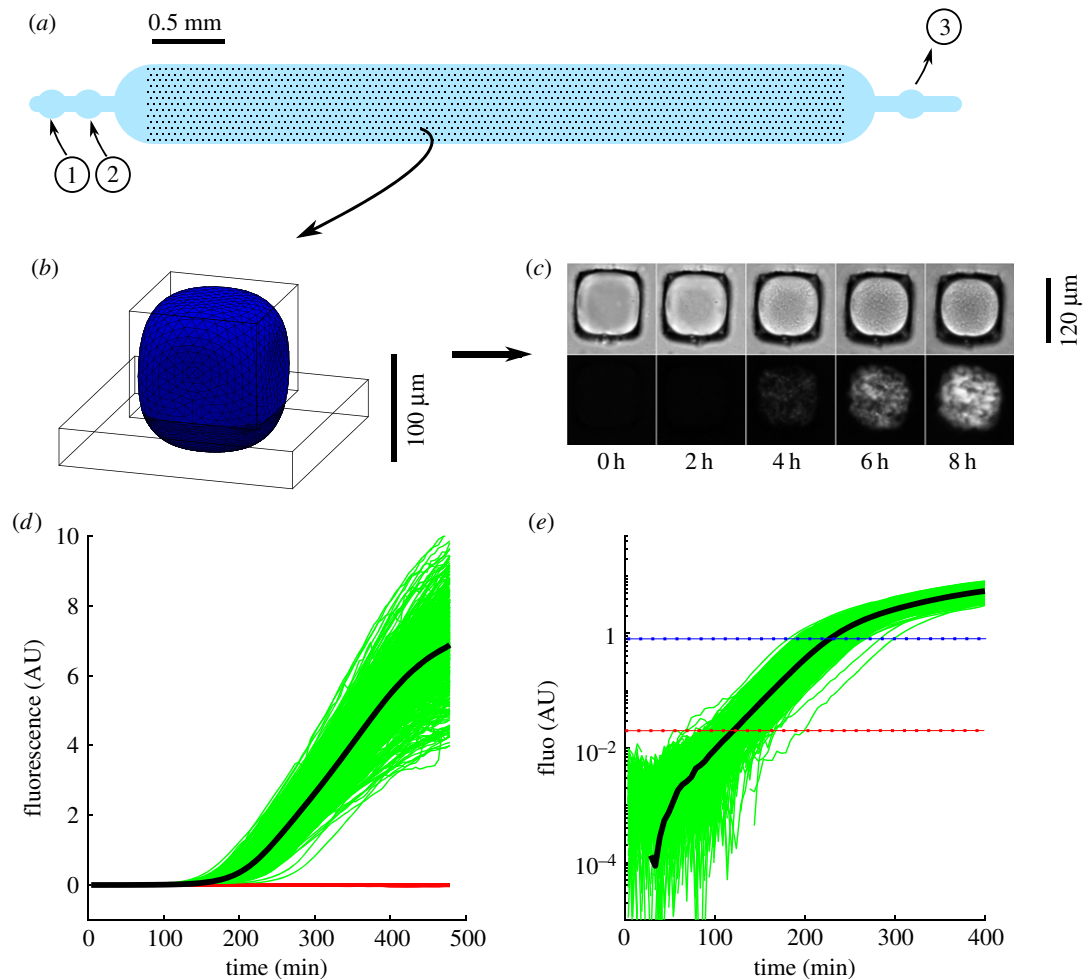


Figure 4. General view of the microfluidic chip and on-chip experiments. (a) Top schematic view of the microfluidic chip, with two inlets and one outlet. Inlet (1) is used to flow an oil solution while inlet (2) is used to flow the aqueous bacteria suspension. The observation chamber comprises 1495 anchors that each contain a single droplet. (b) Perspective view of a microfluidic droplet in an anchor, obtained with Surface Evolver [34]. The aqueous droplet contains culture medium and cells, and is surrounded by an inert oil. (c) Time-lapse in bright-field (top) and fluorescence (bottom) imaging of a droplet that contains bacteria. (d) Fluorescence curves obtained during a growth experiment of *E. coli* at 37°C. Time resolution: 5 min, number of droplets: 915. The green curves are droplets containing bacteria, the red ones are the empty droplets. The black bold line is the mean of bacteria-containing droplets. (e) Same as (d) but plotted on log-lin scale. The bottom dotted line represents the limit of detection, the top one indicates the end of the exponential phase. (Online version in colour.)

measurements display an exponential growth during a time window of nearly 2 h, as shown in figure 4e.

3.2. Single-cell measurements

The experimental data thus generated are similar to data from Monte Carlo simulations, in that they correspond to many repetitions of the same experiment, and enable evaluation of the effect of the cell-to-cell fluctuations. However, to compare experiments and numerical results, simulations need as an input an independent measurement of the division times of the bacteria. These measurements are obtained at the single-cell level, by confining bacteria between a glass slide and a thin agarose pad (see Material and methods and electronic supplementary material, figure S4 for details). We find that the distribution of division times can be well approximated by a Gaussian with mean τ and standard deviation σ , see electronic supplementary material, figure S4(B,C). Moreover, the parameters of the Gaussian distribution evolve with the generation number, a sign of the adaptation of bacteria to their new environment [35]. For *Escherichia coli*, the average generation time decreases from 33 to 21 min between the first and the fourth generation, while the coefficient of

variation $cv_\mu = \sigma/\tau$ decreases from 0.28 to 0.19 over the same time. We find similar results for *Bacillus subtilis*, see electronic supplementary material, figure S4(D), in agreement with results previously reported in the literature on *E. coli* [8,32] and *Salmonella enterica* [31]. Fitting the distribution of division times to a gamma distribution, as suggested in [29,36], does not qualitatively affect our results, see electronic supplementary material, figure S5.

3.3. Comparing microfluidic experiments with theoretical predictions

The distribution of division times from the single-cell experiments can now be used as inputs to obtain theoretical and numerical predictions. The mean and standard deviation of the division times of our strains were experimentally determined for the first four generations, and are considered to be constant after the fourth generation. These can be used as inputs into equations (2.3) and (2.19) to obtain the theoretical growth rate and coefficient of variation, respectively, and as inputs of the Monte Carlo simulations to compute the numerical shape of the PDF. The Poisson parameter is obtained directly from the chip experiment by counting the

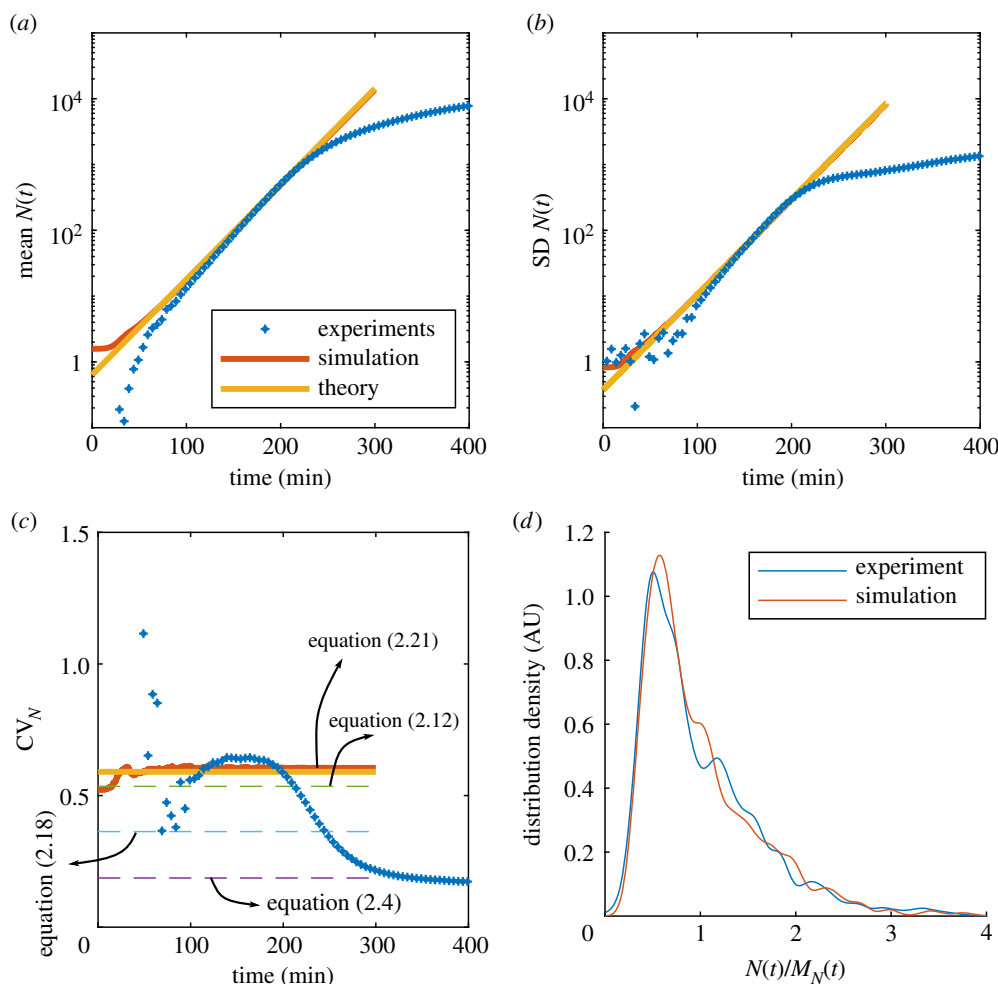


Figure 5. Comparison between the model (yellow), numerical (orange) and experimental results (blue) (same experiment as in figure 4). (a) Mean number of cells as a function of time $M_N(t)$. (b) Standard deviation of the number of cells as a function of time $SD_N(t)$. (c) Coefficient of variation of the number of cells as a function of time cv . Dashed lines correspond to the asymptotic value of cv when the sole source of variability is the cell division time (classical B-H theory), in the presence of an initial Poisson distribution of cells, and in the presence of both a Poisson distribution of cells and an evolving distribution of division times. (d) Shape of the distribution of $N(t)/\exp(\alpha t)$, numerical and experimental, kernel fit. (Online version in colour.)

number of positive droplets. We discuss here the results obtained for *E. coli*. Similar results were obtained for *B. subtilis*, see the electronic supplementary material.

The B-H theory, as well as the classical belief in microbiology [37], predict that the number of bacteria in all droplets should grow exponentially with the same exponential growth rate, at least while nutrients are abundant in the droplet. Experimentally, we recover this exponential growth for all the colonies, see figure 4e, and the mean of the individual growth rates is $\alpha_{\text{exp}} = 2.14 \pm 0.08 \text{ h}^{-1}$, in very good agreement with the theoretical prediction $\alpha_{\text{B-H}} = 2.01 \text{ h}^{-1}$.

All moments of the distribution of the number of cells are predicted to also grow with the same growth rate as the individual curves. The comparisons are shown in figure 5 for the experimental mean and standard deviation. We do observe an exponential growth, and the agreement in terms of growth rate is again very good: we find a growth rate for the mean of $\alpha_{\text{exp}}^{\text{mean}} = 2.19 \text{ h}^{-1}$ and a growth rate for the standard deviation $\alpha_{\text{exp}}^{\text{s.d.}} = 2.22 \text{ h}^{-1}$. Up to the fifth moment of the distribution, we recover an exponential growth experimentally, see the electronic supplementary material.

However the value of this growth rate cannot distinguish between the simple B-H model of §2.2.1 and the complete model of §2.2.4. This distinction can be made by measuring the coefficient of variation CV_N of the number of bacteria,

which is predicted to be constant in the exponential phase by all of the above models. Such a behaviour is indeed observed in the simulations (figure 5c). The experimental coefficient of variation is also approximately constant, but only for a short period of 1 h (see figure 5c). Indeed the exponential growth of the mean and standard deviation makes it difficult to obtain a real plateau: a slight difference in the observed growth rates of these two quantities leads to an exponentially increasing difference between the values of the mean and standard deviation. Yet, the experimental value of the plateau during this period is in good agreement with the expected theoretical value (figure 5c). Note that all sources of variability need to be taken into account to obtain a good agreement between the experimental and theoretical CV_N . Considering solely stochastic cell division, as in the classical B-H model, significantly underestimates the experimental CV_N (dashed purple line). As more sources of variability are added to the model, the coefficient of variation increases. The effect of both the initial Poisson distribution (cyan dashed line), and of the generation-dependent division times (green dashed line) on the value of the coefficient of variation are shown in figure 5c.

Note that the proportionality coefficient between the fluorescent signal and the number of bacteria cancels out when computing the ratio n_2/n_1 . As such raw experimental data

can be directly compared with the simulations and the theory with no fitting parameter. The agreement that is observed between experimental data and theory shows that all sources of variability have to be taken into account to understand the variability in behaviours between droplets: the initial Poisson distribution, the first generation times and the cell-to-cell variability through a B-H process. When one of these sources of variability is not taken into account, the error between experimental results and the theoretical ones is larger, see dashed lines in figure 5c.

Finally, we verify that the shape of the experimental distribution of fluorescence, rescaled by its mean value, converges to a stationary curve (see the electronic supplementary material, movie S3). The convergent behaviour is limited to the exponential phase, before cells enter the stationary phase. Furthermore, the shape of this distribution is in very good agreement with the numerical prediction of the full model, as shown in figure 5d. The distribution is here rescaled by the mean and not by $\exp(\alpha t)$ to get rid of the coefficient of proportionality between the fluorescence and the number of cells. Thus, the distribution of the number of cells in the microfluidic droplet is indeed the one expected from the microscopic cell-to-cell variability.

4. Summary and discussions

In summary, we showed first that the three models of bacterial growth (adder, sizer and timer) were indistinguishable on the scale of even small populations (figure 1). This fact is not surprising since single-cell measurements were required to discriminate between the three models [8]. Realizing this, however, allows us to develop a theoretical description based on the B-H model, which has the advantage of providing several analytical predictions. These predictions are extended here to account for additional sources of variability in the droplet experiments, including a variable number of cells per droplet initially and a division time that varies with the generation number. The analytics are validated with Monte Carlo simulations that show excellent agreement for the different parameters (electronic supplementary material, figure S2).

The four predictions of the expanded B-H model are then verified with microfluidics experiments: (i) the exponential growth in each droplet with a constant exponential growth rate; (ii) the exponential growth of the mean, standard deviation, and higher moments of the cell number distribution, all with the same exponent; (iii) a constant value of the coefficient of variation CV_N that shows a strong dependence on the model details; (iv) a convergence for long durations of the rescaled distribution of population sizes in all droplets. Although the value of the growth rate (2.3) is unmodified compared with the classic B-H model, recovering the shape of the asymptotic distribution and therefore the coefficient of variation requires the full model presented in equations (2.20) and (2.21).

These results are of interest in modelling the growth of microorganisms in their natural environments. Indeed, bacterial cells are often distributed randomly in nearly isolated compartments, e.g. in the soil, in foods, or in local infections. Our results show that the growth of such colonies depends strongly on the initial conditions, particularly on the initial number of cells in the inoculum and on the delay before they reach the steady-state division process. The theoretical description developed here

therefore relates the uncertainty on the initial conditions to the probability distribution of the population size at any instant in time. It thus provides bounds on the expected growth of these semi-isolated populations.

From a more methodological point of view, the droplet measurements present an intermediate scale between single-cell and bulk measurements. Indeed, measuring the distribution of behaviours over many droplets, combined with mathematical modelling, provides a way to access very large statistics on single-cells without the need for tracking the cells individually. This was famously shown by Luria & Delbruck [38] who compared the distribution of mutants in droplets with theoretical predictions in order to understand the origin of mutations. Our analysis shows that the effect of cell-to-cell variability on the distribution of population sizes is dominated by other sources of randomness in the experiments, making it difficult to extract single-cell information from the bulk measurements in the current system. This is quite intuitive, as these other sources of stochasticity come into play at the beginning of the growth: the differences get amplified by the exponential growth afterwards. This can be improved in the future by reducing the uncertainty due to the variable division times (e.g. by synchronizing the cells initially) or from the Poisson distribution (e.g. by working at smaller values of λ). More interestingly, however, the current results provide a baseline for the expected behaviour when the bacterial phenotype is normally distributed. This baseline can then be compared with measurements of growth under stress conditions or by looking at interactions between different strains in order to detect departures from the normal behaviour. This will enable measurements of the heterogeneity of cellular response to external factors, without the need to observe the cells individually.

5. Material and methods

5.1. Bacterial strains and growth media

Bacillus subtilis strain GM2938, a kind gift from Dominique Le Coq (Micalis Institute, INRA, Université Paris-Saclay), has an mCherry fluorescent reporter integrated by double crossing in *amyE*, and a resistance gene for spectinomycin [*amyE::PhyperSpank-mCherry* "Rudner", *lacI*/SpecR]. Fluorescence was induced by the addition of IPTG (SIGMA-ALDRICH, St Louis, Missouri). All growth media used for this strain contain IPTG at a concentration of 1 mM. The strain *E. coli* AV04 was a gift from Antoine Vigouroux at Institut Pasteur (Paris, France) and constitutively expresses an mCherry fluorescent reporter [39]. Bacteria are grown in rich LB medium (SIGMA-ALDRICH L3022), supplemented with IPTG at a concentration of 1 mM in the case of *B. subtilis*. For all experiments, bacteria are grown overnight in fresh liquid medium from an LB-agar plate, then re-diluted at 1/200 in fresh medium until they reach an $OD_{600} \sim 0.4$ (Amersham, Little Chalfont, UK). Bacteria are then used for the on-chip or single-cell experiments. For *B. subtilis*, we also added spectinomycin (SIGMA) at 0.1 mg ml^{-1} in the overnight culture to avoid contamination. All culturing steps were conducted at 37°C .

5.2. Monitoring individual division times

The division times of individual bacteria were measured by imaging under a Nikon Ti phase contrast microscope, at a $90\times$ magnification ($60\times$ objective and 1.5 optical zoom). A bacterial suspension with an $OD_{600} \sim 0.4$ was centrifuged and diluted in fresh LB 2X to obtain an OD_{600} of ~ 0.2 , suitable for individual bacteria observation. For *B. subtilis*, the bacterial suspension

was mixed in a 1:1 ratio with ultra-low gelling temperature agarose (3%, SIGMA A2576). Then, 1 μ l of the resulting mixture was deposited on a glass slide, and covered with a thin (0.5 cm) PDMS slab to allow for oxygen diffusion. The sample was left at room temperature for 15 min to let the agarose gelify, then placed on a heating stage (Warner Instruments) at 37°C, and monitored with a phase-contrast microscope. Ten positions containing initially 1–3 bacteria were chosen and scanned for 3 h every 4 min. The division times of all bacteria were manually determined. For *E. coli*, 0.5 μ l of bacteria at OD₆₀₀ ~ 0.2 were sandwiched between a glass slide and a homemade 50/50 LB/3% agarose pad, and then observed under the microscope every 3 min for 3 h. Division times were determined with a segmentation algorithm [40].

5.3. Microfluidic experiments

The microfluidic chip used in all experiments consists of a chamber of 45 × 5 mm containing 1495 square surface energy traps, or anchors, as described in [30] (figure 4a). A bacterial suspension with an OD₆₀₀ ~ 0.4 is loaded at room temperature, and 2 nl droplets are created *in situ* on surface-energy anchors [41] by pushing the suspension with oil [30] (figure 4b). The whole process takes approximately 15 min. We verified that the initial number of bacteria per droplet follows a Poisson distribution, see [30]. The chip is then placed on the motorized stage of an epi-fluorescence Nikon Ti microscope, with a temperature control at 37°C. Fluorescence images of the droplets are taken at a 10× magnification with an EMCCD Camera (Andor Technologies, Belfast, UK). About 800 droplets can be scanned in 5 min. Droplets are scanned every 5 min during 8 h, after which bacteria stop growing, see figure 4c for images. Then, droplets are scanned once per hour

for 5 h to monitor the final distribution of fluorescence intensity. Bacterial growth can be monitored by integrating the fluorescence signal of each trap at each time step [30,42], see figure 4d for the experimental growth curves. The intensity of the fluorescent signal is proportional to $N(t)$, the number of cells in the droplet, with a measurement noise that is independent of the number of cells, see the electronic supplementary material for more details. Because of the initial Poisson distribution, some droplets are empty. Their fluorescence (red lines in figure 4d) is used to quantify the measurement noise (see the electronic supplementary material). For droplets that contain bacteria, we define the detection and saturation limits manually and once and for all the droplets. Between these two limits the growth rate is obtained for each droplet by fitting the growth curve with an exponential.

Data accessibility. This article has no additional data.

Authors' contributions. C.B., G.A. and A.B. conceived the study. A.B. performed the experiments, the simulations and the theoretical computations. M.S.S.J. performed the single-cell experiments with *B. subtilis*. C.B., G.A. and A.B. wrote the manuscript.

Competing interests. We declare we have no competing interests.

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